Rohan Desai

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PROFILE

Full-stack and AI engineer, currently SDE-2 at Amber, a global student-housing marketplace. Built systems the business runs on: an LLM service that turns 6M+ customer conversations into instant lead intelligence, the engine that routes every incoming lead to the right salesperson, and the platform teams use to automate their workflows. Comfortable owning a system end to end, from data ingestion and model pipelines to the product on top.

PROFESSIONAL EXPERIENCE

Software Development Engineer 2, AmberStudent May 2024 – Present

AI & LLM systems

- Designed and built Prism (Lead Memory), an AI service that turns 6.2M+ communications across 177K+ leads (WhatsApp, calls, emails, SMS, notes) into a single profile an agent can read in 30 seconds. Kinesis CDC streams changes from Postgres into DocumentDB, a map-reduce LLM pipeline with debounced refresh workers keeps profiles current, and the CRM reads them over a token-authenticated Node/Hono API (TypeScript).
- Built the CRM screens that surface these profiles: lead facts, story, risks, suggested next steps, and coaching notes, with evidence links and a paginated communication timeline (React).
- Extended Prism into an automated sales-audit pipeline, replacing a team's manual call reviews. An LLM judge scores every call and chat transcript against a versioned, hash-locked 17-parameter rubric and must cite quotes from the transcript as evidence; all scoring arithmetic stays in TypeScript, so the model can never invent a grade, and missing data is excluded and its weight redistributed rather than scored as failure.
- Rolled those audits up into per-salesperson report cards: sampled won and lost leads, parameter-level aggregation, and an LLM-written narrative of each person's patterns, giving managers continuous performance feedback instead of spot checks.
- Shipped an image-quality pipeline for property listings that catches blurred uploads (about 1 in 60) and restores them automatically with OpenAI image-edit models, plus background upscaling workers (Rails, Sidekiq).

Lead assignment & CRM platform

- Own the lead-distribution engine (DCM): conversion-weighted agent scoring with Bayesian smoothing, rule-based bucket matching, per-agent region filters, round-robin modes, and workers that catch and reassign misrouted or stale leads.
- Core engineer on the CRM re-platform: set up the new React 19 + Vite + TanStack app and rebuilt inventory onboarding, bulk commission management, and supply-task dashboards on it.
- Worked on the migration from a fixed 3-level to a dynamic n-level inventory hierarchy: v3 inventory APIs that read from both old and new structures, with OpenSearch-backed sellable-unit search.
- Built the partner commissions system: country-level and unit-type commission definitions, bulk creation across inventories, and automated recalculation.

Automation, integrations & security

- Extended an open-source AI workflow builder into the CRM's production automation platform (Next.js, React Flow, AWS SQS): event-driven execution, database-driven cron scheduling, custom JavaScript/TypeScript code steps, and CRM email integration.
- Built self-serve bulk WhatsApp messaging for sales campaigns: upload a CSV, preview, send, and get per-message delivery reports with failure reasons. Sales no longer needs an engineer for campaign sends.
- Looked after telephony and messaging integrations: Plivo dialer revamp with phone-number rotation and PSTN controls, WhatsApp platforms (Eazybe, DoubleTick), Trustpilot reviews, and Slack/Suptask ticketing sync.
- Locked down PII access on internal APIs: role-based ACLs on sensitive endpoints, per-user rate limiting (30 req/min, automatic 3-hour blocks) with Slack alerts, and encrypted identifiers.
- Performance and reliability work: a Ruby memory-fragmentation fix rolled out across all service Dockerfiles, database indexes and query fixes for lead notes and emails, list virtualization, and a mobile-responsive CRM.

Junior Software Engineer, Torinit Technologies Inc. 2022 – 2024

Right at Home Realty (real estate)

- Engineered scalable microservices on AWS Lambda and Node.js, keeping the platform highly available.
- Deployed and managed APIs with the Serverless framework.
- Cut response times of existing REST APIs by up to 50% through caching and parallelization.

Exo (automobiles)

- Worked as a full-stack developer on a remote, distributed, fast-paced team.
- Used Cloudflare Workers for edge deployment, improving global application performance and reducing latency.
- Brought page load times down to 200–500ms by fetching JavaScript dynamically over WebSockets.

SKILLS

Languages & Frameworks: TypeScript, JavaScript, Ruby on Rails, Node.js, React (18/19), Next.js, Redux, TanStack, Hono, REST APIs

AI & LLM: LLM summarization pipelines (map-reduce), prompt engineering, transcript scoring & evaluation, OpenAI image models, Vercel AI SDK, workflow automation (React Flow)

Data & Infra: PostgreSQL, MongoDB / AWS DocumentDB, Redis, OpenSearch, AWS (Kinesis, ECS, Lambda, S3), Docker, Sidekiq, CDC / event-driven pipelines, CI/CD

Integrations & Observability: Plivo, WhatsApp Business APIs, Slack, Datadog, Grafana, Metabase, Rollbar, LogRocket

EDUCATION

B.E. Information Technology, University of Mumbai 2018 – 2022